Question 1: Basic PHP Syntax and Functions (4 Points)

1.1 (2 points) Write a PHP script that declares three variables:

- yourFullName (your full name as a string)
- yourBirthYear (your birth year as an integer)
- currentYear (current year as an integer – hard-code 2025)

Then calculate your age and output a sentence in the following format: "My name is [yourFullName], I was born in [yourBirthYear], so I am [age] years old in 2025."

1.2 (2 points) Create a PHP function named convertTemperature that takes two parameters:

- celsius (float) – temperature in Celsius
- toFahrenheit (boolean) – true = convert to Fahrenheit, false = convert to Kelvin

The function should return:

- Fahrenheit if toFahrenheit = true: $F = C \times 9/5 + 32$
- Kelvin if toFahrenheit = false: $K = C + 273.15$

Demonstrate the function twice:

- Convert 25°C to Fahrenheit
- Convert 25°C to Kelvin

Output example: "25°C = 77°F" "25°C = 298.15 K"

Question 2: Arrays and Control Structures (4 Points)

2.1 (2 points) Create an indexed array named $vietnamRivers containing at least 6 famous Vietnamese rivers. Use a **while** loop to display each river name as an HTML unordered list item (<ul><li>...</li></ul>).

Example output:

text

```
<ul>
    <li>Red River</li>
    <li>Mekong River</li>
    ...
</ul>
```

2.2 (2 points) Given the following associative array:

PHP

```php
$exchange2025 = [
    'USD' => 25200,
    'EUR' => 27200,
    'SGD' => 18800,
    'AUD' => 16800
];
```

Write a PHP script that:

- Increases every exchange rate by 3.5% (rounded to nearest integer)
- Outputs the updated rates in an HTML definition list (<dl><dt>...</dt><dd>...</dd></dl>)

Example output:

text

```text
<dl>
    <dt>USD</dt><dd>26,082 VND</dd>
    <dt>EUR</dt><dd>28,152 VND</dd>
    ...
</dl>
```

Question 3: Class & Object – Book Management (6 Points)

3.1 Create Book Class (3 Points) Create a class named Book with the following members:

- Private properties: $isbn, $title, $author, $price, $publishYear
- Constructor that accepts all 5 parameters
- Getters for all properties
- A method getAge() that returns how old the book is in 2025 (2025 - publishYear)
- A method isExpensive() that returns true if price > 500000

3.2 Array of Book Objects (3 Points) Create an indexed array named $library containing **4 Book objects** with realistic data.

Display all books in an HTML table with columns: ISBN | Title | Author | Price (VND) | Publish Year | Book Age | Expensive?

Use the getAge() and isExpensive() methods in the table.

UI Mockup:

text

```
-------------------------------------------------------
| Library Books                                       |
-------------------------------------------------------
| ISBN       | Title              | Author        | Price     | Year | Age |
Expensive? |
|------------|--------------------|---------------|-----------|------|-----|-
-----------|
| 978-604... | To Kill a Mockingbird | Harper Lee | 320,000   | 1960 | 65  |
No         |
| ...        | ...                | ...           | ...       | ...  | ... |
...        |
-------------------------------------------------------
```

Question 4: Product Management with Form & Session (6 Points)

Objective: Create and manage Product objects with user input and session.

4.1 Create Product Class (3 Points) Create a class named Product with:

- Private properties: $id, $name, $price, $quantity, $category
- Constructor
- Getters and Setters for all properties
- Method getTotalValue() that returns price × quantity
- Method getStatus() that returns:
  - "Available" if quantity ≥ 5
  - "Low Stock" if 1 ≤ quantity < 5
  - "Out of Stock" if quantity == 0

4.2 Form to Add Product & Display List (3 Points) Create an HTML form that allows the user to input:

- Product ID (text)
- Product Name (text)
- Price (number)
- Quantity (number)
- Category (select: Electronics, Clothing, Food, Others)

When the form is submitted (to the same page):

- Create a new Product object
- Store the object in $_SESSION['productList'] (array of objects)
- Display a success message: "Product [name] (ID: [id]) has been added successfully!"

Below the form, display all products from the session array in a table. Show the result of getTotalValue() and getStatus() in extra columns.

UI Mockup:

text

```
-------------------------------------------------------
| Add New Product                                     |
-------------------------------------------------------
| ID: [________]                                      |
| Name: [_________________________]                   |
| Price: [________] VND                               |
| Quantity: [________]                                |
| Category: [_________________________]               |
| [ Add Product ]                                     |
-------------------------------------------------------


Current Products:
-------------------------------------------------------
| ID   | Name          | Price    | Qty | Category   | Total Value | Status
|
| P001 | Smart TV      | 12,000,000| 4  | Electronics| 48,000,000  | Low
Stock  |
-------------------------------------------------------
```

Question 5: Form Handling, Data Validation & File Upload (4 Points)

5.1 (2 points) Create an HTML form that allows a user to input the following information:

- Full Name (text)
- Phone Number (text)
- Profile Picture (file upload – only jpg, jpeg, png allowed, max 3MB)
- Submit button ("Submit Profile")

The form should submit to the same PHP page using **POST** method and **enctype="multipart/form-data"**.

5.2 (2 points) On form submission, perform the following validations and actions:

- Full Name must not be empty and must contain at least 6 characters.

- Phone Number must match Vietnamese mobile format (starts with 09/08/03/07..., 10 digits).
- Profile picture must be uploaded, must be jpg/jpeg/png, and size ≤ 3MB.
- If all validations pass, save the uploaded file to a folder named profiles/ with a unique name (e.g., timestamp_username.jpg).
- Display a success message including the uploaded file name and a preview image (<img src="profiles/filename.jpg" width="200">).
- If any validation fails, display appropriate error message(s) in red.

Example success message:

text

```
Profile submitted successfully!
Full Name: Tran Thi Kim Anh
Phone: 0987654321
Uploaded photo: 1712345678_kimanh.jpg
[Image preview here]
```

End of Exam